

Prompts for the LLM-Assisted Pharmacovigilance Pipeline

1. Prompt 1: This prompt was used in the Step 2.

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SYSTEM PROMPT

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You are a senior pharmacovigilance analyst and MedDRA expert.

Definitions:

- Adverse Event (AE): any untoward medical occurrence in a patient administered a medicinal product, whether or not causally related to treatment.
- Adverse Drug Reaction (ADR): a noxious and unintended response for which a causal relationship with the medicinal product is at least a reasonable possibility.
- MedDRA Preferred Term (PT): one coded medical concept recorded in a FAERS report.

Background:

I have a list of PTs obtained from FAERS reports. FAERS records reported adverse events, which are not necessarily adverse drug reactions or causally related to a drug. I want to identify PTs that could plausibly represent an adverse drug reaction to some drug.

Based on an initial review of the PTs, I created the following categories to help determine which PTs could plausibly represent ADRs.

Use only these categories:

- CLINICAL_EVENT: A genuine sign, symptom, diagnosis, or injury experienced by the patient (e.g. Pancreatitis, Rash, Seizure, Hepatic failure).
- LAB_OR_INVESTIGATION: A laboratory test, medical imaging, or vital-sign measurement, including a change or abnormality in its result (e.g. Blood glucose increased, Liver function test abnormal).
- LACK_OF_EFFICACY: Treatment failure, recurrence due to lack of effect, or worsening/progression of the condition being treated (e.g. Drug ineffective, Condition aggravated).
- MEDICATION_ERROR_OR_MISUSE: A medication error, accidental or intentional misuse, abuse, overdose, underdose, inappropriate use, or incorrect administration (e.g. Accidental overdose, Off label use).
- PRODUCT_OR_DEVICE_ISSUE: A quality complaint, malfunction, defect, packaging issue, or other problem involving the medicinal product or device itself, without specifying a patient injury or clinical reaction (e.g. Device malfunction, Product quality issue).
- ADMINISTRATION_SITE_OR_ROUTE: A clinical reaction specifically localized to an administration, infusion, injection, application, implantation, or instillation site (e.g. Injection site pain, Extravasation).
- NONSPECIFIC_OR_CONTEXT: A concept that is not intrinsically a patient clinical event and does not fit another category, including procedures, treatment or route context, drug-level monitoring, hospitalization context, administrative terms, or explicit no-event terms, and vague non-specific descriptors (e.g. Drug level increased, Dialysis, Drug intolerance, No adverse event).

Goal:

Classify each PT into exactly one of the categories above according to what the PT intrinsically represents.

Classification rules:

1. Make the classification drug-agnostic. Do not consider whether the PT is associated with or plausible for a particular drug.
2. Classify the PT itself, not a possible cause, consequence, or associated condition.

3. If a PT could fit multiple categories, select the category that most directly describes the PT's primary MedDRA concept.
4. If none of the categories clearly applies, use NONSPECIFIC_OR_CONTEXT.
5. Include every input PT exactly once and preserve the original order and spelling.

Respond only with valid JSON in this exact structure:

```
{"results":[{"pt":"<verbatim>","pt_type":"<CATEGORY>"}]}
```

Do not include explanations, Markdown, comments, additional keys, or text outside the JSON.

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USER PROMPT

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Classify the following 50 PTs:

1. Drug ineffective
2. Toxicity to various agents
3. Acute kidney injury
4. Hypotension
5. Drug interaction
6. Treatment failure
7. Drug ineffective for unapproved indication
8. Foetal exposure during pregnancy
9. Maternal exposure during pregnancy
10. Intentional overdose
11. Somnolence
12. Anaemia

13. Thrombocytopenia
14. Overdose
15. Pyrexia
16. Tachycardia
17. Completed suicide
18. Hepatic cytolysis
19. Premature baby
20. Neutropenia
21. Multiple organ dysfunction syndrome
22. Suicide attempt
23. Therapy non-responder
24. Disease progression
25. Drug resistance
26. Drug abuse
27. Off label use
28. Cardiac arrest
29. Electrocardiogram QT prolonged
30. Hypokalaemia
31. Drug intolerance
32. Bradycardia
33. Respiratory failure
34. Vomiting
35. Febrile neutropenia
36. Diarrhoea
37. Hyponatraemia

- 38. Septic shock
- 39. Pancytopenia
- 40. Leukopenia
- 41. Dyspnoea
- 42. Agranulocytosis
- 43. Exposure during pregnancy
- 44. Confusional state
- 45. Drug hypersensitivity
- 46. Nausea
- 47. Agitation
- 48. Depression
- 49. Therapeutic product effect incomplete
- 50. Coma

2. Prompt 2: This prompt was used in the Step 3. This is a long prompt due to drug label information. This is an example of the drug, "FEBUXOSTAT."

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SYSTEM PROMPT

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You are a senior pharmacovigilance expert comparing candidate PTs for one drug against that drug's official FDA label.

Definitions:

- Adverse Drug Reaction (ADR): a noxious and unintended response for which a causal relationship with the medicinal product is at least a reasonable possibility.
- MedDRA Preferred Term (PT): one coded medical concept recorded in a FAERS report.

Background:

The label text is provided as reference material and may include any of the drug's sections -- indications, boxed warning, adverse reactions (narrative text and incidence tables), warnings, precautions, and contraindications. Treat the ENTIRE label as the source of truth; do not limit your search to any single section.

Classification rules:

For each candidate PT, make a hierarchical decision and assign exactly ONE status:

1. If the PT is really the drug's INDICATION, the underlying disease being treated, or that disease worsening / the drug failing to work (so it's confounding or treatment failure, not a reaction) -> DISEASE_RELATED.
2. Else if the label describes the PT as something the drug can cause -- a risk, warning, or adverse experience of taking it -- in ANY section -> ALREADY_LABELED. A bare mention is NOT enough: a condition named only as a pre-existing risk factor or a contraindicated patient population (e.g. "contraindicated in patients with hepatic impairment") is not ALREADY_LABELED.

3. Else -> NOVEL (a potential unlabeled ADR).

Respond only with valid JSON in this exact structure:

```
{"results":[{"pt":"<verbatim>","status":"<DISEASE_RELATED|ALREADY_LABELED|NOVEL>","reasoning":"<one short sentence>"}]}
```

Do not include Markdown, comments, additional keys, the label text, or any text outside the JSON.

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USER PROMPT (16,596 chars)

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DRUG (active ingredient): FEBUXOSTAT

FDA LABEL:

INDICATIONS AND USAGE

1 INDICATIONS AND USAGE ULORIC is a xanthine oxidase (XO) inhibitor indicated for the chronic management of hyperuricemia in adult patients with gout who have an inadequate response to a maximally titrated dose of allopurinol, who are intolerant to allopurinol, or for whom treatment with allopurinol is not advisable. ULORIC is a xanthine oxidase (XO) inhibitor indicated for the chronic management of hyperuricemia in adult patients with gout who have an inadequate response to a maximally titrated dose of allopurinol, who are intolerant to allopurinol, or for whom treatment with allopurinol is not advisable. (1) Limitations of Use : ULORIC is not recommended for the treatment of asymptomatic hyperuricemia. (1) Limitations of Use : ULORIC is not recommended for the treatment of asymptomatic hyperuricemia.

BOXED WARNING

WARNING: CARDIOVASCULAR DEATH Gout patients with established cardiovascular (CV) disease treated with ULORIC had a higher rate of CV death compared to those treated with allopurinol in a CV outcomes study [see Warnings and Precautions (5.1)] . Consider the risks and benefits of ULORIC when deciding to prescribe or continue patients on ULORIC. ULORIC should only be

used in patients who have an inadequate response to a maximally titrated dose of allopurinol, who are intolerant to allopurinol, or for whom treatment with allopurinol is not advisable [see Indications and Usage (1)] . WARNING: CARDIOVASCULAR DEATH See full prescribing information for complete boxed warning. Gout patients with established cardiovascular (CV) disease treated with ULORIC had a higher rate of CV death compared to those treated with allopurinol in a CV outcomes study. (5.1) Consider the risks and benefits of ULORIC when deciding to prescribe or continue patients on ULORIC. ULORIC should only be used in patients who have an inadequate response to a maximally titrated dose of allopurinol, who are intolerant to allopurinol, or for whom treatment with allopurinol is not advisable. (1)

ADVERSE REACTIONS

6 ADVERSE REACTIONS The following serious adverse reactions are described elsewhere in the prescribing information: Cardiovascular Death [see Warnings and Precautions (5.1)] Hepatic Effects [see Warnings and Precautions (5.3)] Serious Skin Reactions [see Warnings and Precautions (5.4)] Adverse reactions in $\geq 1\%$ of patients treated with ULORIC are liver function abnormalities, nausea, arthralgia, and rash. (6.1) To report SUSPECTED ADVERSE REACTIONS, contact Takeda Pharmaceuticals America, Inc. at 1-877-TAKEDA-7 (1-877-825-3327) or FDA at 1-800-FDA-1088 or www.fda.gov/medwatch. 6.1 Clinical Trials Experience Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in practice. In Phase 2 and 3 clinical studies, a total of 2757 patients with hyperuricemia and gout were treated with ULORIC 40 mg or 80 mg daily. For ULORIC 40 mg, 559 patients were treated for ≥ 6 months. For ULORIC 80 mg, 1377 patients were treated for ≥ 6 months, 674 patients were treated for ≥ 1 year and 515 patients were treated for ≥ 2 years. In the CARES study, a total of 3098 patients were treated with ULORIC 40 mg or 80 mg daily; of these, 2155 patients were treated for ≥ 1 year and 1539 were treated for ≥ 2 years [see Clinical Studies (14.2)] . Most Common Adverse Reactions In three randomized, controlled clinical studies (Studies 1, 2 and 3), which were 6 to 12 months in duration, the following adverse reactions were reported by the treating physician as related to study drug. Table 1 summarizes adverse reactions reported at a rate of at least 1% in ULORIC treatment groups and at least 0.5% greater than placebo. Table 1: Adverse Reactions Occurring in $\geq 1\%$ of Patients Treated with ULORIC and at Least 0.5% Greater than in Patients Receiving Placebo in Controlled Studies

Placebo	ULORIC	allopurinol
Of the patients who received allopurinol, 10 received 100		

mg, 145 received 200 mg, and 1122 received 300 mg, based on level of renal impairment.

Adverse Reactions (N=134) 40 mg daily (N=757) 80 mg daily (N=1279) (N=1277) Liver Function Abnormalities 0.7% 6.6% 4.6% 4.2% Nausea 0.7% 1.1% 1.3% 0.8% Arthralgia 0% 1.1% 0.7% 0.7% Rash 0.7% 0.5% 1.6% 1.6% The most common adverse reaction leading to discontinuation from therapy was liver function abnormalities in 1.8% of ULORIC 40 mg, 1.2% of ULORIC 80 mg, and in 0.9% of patients treated with allopurinol. In addition to the adverse reactions presented in Table 1, dizziness was reported in more than 1% of patients treated with ULORIC although not at a rate more than 0.5% greater than placebo. In the CARES study, liver function abnormalities and diarrhea were reported in more than 1% of patients treated with ULORIC, although not at a rate more than 0.5% greater than allopurinol.

Less Common Adverse Reactions In clinical studies the following adverse reactions occurred in less than 1% of patients and in more than one subject treated with doses ranging from 40 mg to 240 mg of ULORIC. This list also includes adverse reactions (less than 1% of patients) associated with organ systems from Warnings and Precautions.

Blood and Lymphatic System Disorders: anemia, idiopathic thrombocytopenic purpura, leukocytosis/leukopenia, neutropenia, pancytopenia, splenomegaly, thrombocytopenia. Cardiac Disorders: angina pectoris, atrial fibrillation/flutter, cardiac murmur, ECG abnormal, palpitations, sinus bradycardia, tachycardia. Ear and Labyrinth Disorders: deafness, tinnitus, vertigo. Eye Disorders: vision blurred. Gastrointestinal Disorders: abdominal distention, abdominal pain, constipation, dry mouth, dyspepsia, flatulence, frequent stools, gastritis, gastroesophageal reflux disease, gastrointestinal discomfort, gingival pain, hematemesis, hyperchlorhydria, hematochezia, mouth ulceration, pancreatitis, peptic ulcer, vomiting. General Disorders and Administration Site Conditions: asthenia, chest pain/discomfort, edema, fatigue, feeling abnormal, gait disturbance, influenza-like symptoms, mass, pain, thirst. Hepatobiliary Disorders: cholelithiasis/cholecystitis, hepatic steatosis, hepatitis, hepatomegaly. Immune System Disorder: hypersensitivity. Infections and Infestations: herpes zoster. Procedural Complications: contusion. Metabolism and Nutrition Disorders: anorexia, appetite decreased/increased, dehydration, diabetes mellitus, hypercholesterolemia, hyperglycemia, hyperlipidemia, hypertriglyceridemia, hypokalemia, weight decreased/increased. Musculoskeletal and Connective Tissue Disorders: arthritis, joint stiffness, joint swelling, muscle spasms/twitching/tightness/weakness, musculoskeletal pain/stiffness, myalgia. Nervous System Disorders: altered taste, balance disorder, cerebrovascular accident, Guillain-Barré syndrome, headache, hemiparesis, hypoesthesia, hyposmia, lacunar infarction, lethargy, mental impairment, migraine, paresthesia, somnolence, transient ischemic attack, tremor. Psychiatric Disorders: agitation, anxiety, depression, insomnia, irritability, libido decreased, nervousness, panic attack, personality change. Renal and Urinary Disorders: hematuria, nephrolithiasis, pollakiuria, proteinuria, renal failure, renal insufficiency, urgency, incontinence. Reproductive System and Breast Changes: breast pain, erectile dysfunction,

gynecomastia. Respiratory, Thoracic and Mediastinal Disorders: bronchitis, cough, dyspnea, epistaxis, nasal dryness, paranasal sinus hypersecretion, pharyngeal edema, respiratory tract congestion, sneezing, throat irritation, upper respiratory tract infection. Skin and Subcutaneous Tissue Disorders: alopecia, angio edema, dermatitis, dermographism, ecchymosis, eczema, hair color changes, hair growth abnormal, hyperhidrosis, peeling skin, petechiae, photosensitivity, pruritus, purpura, skin discoloration/altered pigmentation, skin lesion, skin odor abnormal, urticaria. Vascular Disorders: flushing, hot flush, hypertension, hypotension. Laboratory Parameters: activated partial thromboplastin time prolonged, creatine increased, bicarbonate decreased, sodium increased, EEG abnormal, glucose increased, cholesterol increased, triglycerides increased, amylase increased, potassium increased, TSH increased, platelet count decreased, hematocrit decreased, hemoglobin decreased, MCV increased, RBC decreased, creatinine increased, blood urea increased, BUN/creatinine ratio increased, creatine phosphokinase (CPK) increased, alkaline phosphatase increased, LDH increased, PSA increased, urine output increased/decreased, lymphocyte count decreased, neutrophil count decreased, WBC increased/decreased, coagulation test abnormal, low density lipoprotein (LDL) increased, prothrombin time prolonged, urinary casts, urine positive for white blood cells and protein.

6.2 Postmarketing Experience The following adverse reactions have been identified during post approval use of ULORIC. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure. Blood and Lymphatic System Disorders: agranulocytosis, eosinophilia. Hepatobiliary Disorders: hepatic failure (some fatal), jaundice, serious cases of abnormal liver function test results, liver disorder. Immune System Disorders: anaphylaxis, anaphylactic reaction. Musculoskeletal and Connective Tissue Disorders: rhabdomyolysis. Psychiatric Disorders: psychotic behavior including aggressive thoughts. Renal and Urinary Disorders: tubulointerstitial nephritis. Skin and Subcutaneous Tissue Disorders: generalized rash, Stevens-Johnson Syndrome, hypersensitivity skin reactions, erythema multiforme, drug reaction with eosinophilia and systemic symptoms, toxic epidermal necrolysis.

ADVERSE REACTIONS TABLE

Table 1: Adverse Reactions Occurring in $\geq 1\%$ of Patients Treated with ULORIC and at Least 0.5% Greater than in Patients Receiving Placebo in Controlled Studies

Placebo | ULORIC | allopurinol
Of the patients who received allopurinol, 10 received 100 mg, 145 received 200 mg, and 1122 received 300 mg, based on level of renal impairment.

Adverse Reactions | (N=134) | 40 mg daily (N=757) | 80 mg daily (N=1279) | (N=1277)

Liver Function Abnormalities | 0.7% | 6.6% | 4.6% | 4.2%

Nausea | 0.7% | 1.1% | 1.3% | 0.8%

Arthralgia | 0% | 1.1% | 0.7% | 0.7%

Rash | 0.7% | 0.5% | 1.6% | 1.6%

WARNINGS AND CAUTIONS

5 WARNINGS AND PRECAUTIONS

Gout Flares : An increase in gout flares is frequently observed after initiation of ULORIC. If a gout flare occurs during treatment, ULORIC need not be discontinued. Prophylactic therapy (i.e., non-steroidal anti-inflammatory drug or colchicine) upon initiation of treatment may be beneficial for up to six months. (2.4 , 5.2)

Hepatic Effects : Cases of hepatic failure, some fatal, have been reported. If liver injury is detected, promptly interrupt ULORIC and treat cause, if possible, to resolution or stabilization. Permanently discontinue ULORIC if liver injury is confirmed, and no alternate etiology can be found. (5.3)

Serious Skin Reactions : Serious skin and hypersensitivity reactions, including Stevens-Johnson Syndrome, drug reaction with eosinophilia and systemic symptoms and toxic epidermal necrolysis have been reported in patients taking ULORIC. Discontinue ULORIC if serious skin reactions are suspected. (5.4)

5.1 Cardiovascular Death In a cardiovascular (CV) outcome study, gout patients with established CV disease treated with ULORIC had a higher rate of CV death compared to those treated with allopurinol. Sudden cardiac death was the most common cause of adjudicated CV deaths, 2.7% in the ULORIC group (83 of 3,098) as compared to 1.8% in the allopurinol group (56 of 3,092). ULORIC was similar to allopurinol for nonfatal myocardial infarction (MI), nonfatal stroke and unstable angina with urgent coronary revascularization [see Clinical Studies (14.2)] . Because of the increased risk of CV death, ULORIC should only be used in patients who have an inadequate response to a maximally titrated dose of allopurinol, who are intolerant to allopurinol, or for whom treatment with allopurinol is not advisable [see Indications and Usage(1)] . Consider the risks and benefits of ULORIC when deciding to prescribe or continue patients on ULORIC. Consider use of prophylactic low-dose aspirin therapy in patients with a history of CV disease. Monitor patients for the development of CV events. Inform patients about the symptoms of serious CV events and the steps to take if they occur.

5.2 Gout Flares After initiation of ULORIC, an increase in gout flares is frequently observed. This increase is due to reduction in serum uric acid levels, resulting in mobilization of urate from tissue deposits. In order to prevent gout flares when ULORIC is initiated, concurrent prophylactic treatment with an NSAID or colchicine is recommended [see Dosage and Administration (2.4)] .

5.3 Hepatic Effects Cases of fatal and nonfatal hepatic failure in patients

taking ULORIC have been reported. During randomized controlled studies, transaminase elevations greater than three times the upper limit of normal (ULN) were observed (AST: 2%, 2%, and ALT: 3%, 2% in ULORIC and allopurinol-treated patients, respectively). No dose-effect relationship for these transaminase elevations was noted [see Clinical Pharmacology (12.3)] . Obtain a liver test panel (serum alanine aminotransferase [ALT], aspartate aminotransferase [AST], alkaline phosphatase, and total bilirubin) as a baseline before initiating ULORIC. Measure liver tests promptly in patients who report symptoms that may indicate liver injury, including fatigue, anorexia, right upper abdominal discomfort, dark urine or jaundice. In this clinical context, if the patient presents abnormal liver tests (ALT or AST greater than three times the upper limit of the reference range), interrupt ULORIC treatment while investigating the probable cause. Permanently discontinue ULORIC if liver injury is confirmed, and no alternate etiology can be found. Permanently discontinue ULORIC in patients who have serum ALT or AST greater than three times the reference range with serum total bilirubin greater than two times the reference range without alternative etiologies because they are at risk for severe drug-induced liver injury. For patients with lesser elevations of serum ALT or bilirubin and with an alternate probable cause, treatment with ULORIC can be used with close monitoring. 5.4 Serious Skin Reactions Serious skin and hypersensitivity reactions, including Stevens-Johnson Syndrome, drug reaction with eosinophilia and systemic symptoms (DRESS) and toxic epidermal necrolysis (TEN) have been reported postmarketing in patients taking ULORIC. Discontinue ULORIC if serious skin reactions are suspected [see Patient Counseling Information (17)] . Many of these patients had reported previous similar skin reactions to allopurinol. ULORIC should be used with close monitoring in these patients.

CONTRAINDICATIONS

4 CONTRAINDICATIONS ULORIC is contraindicated in patients being treated with azathioprine or mercaptopurine [see Drug Interactions (7)] . ULORIC is contraindicated in patients being treated with azathioprine or mercaptopurine. (4)

Classify the following 44 PTs:

1. Cerebral artery thrombosis
2. Gouty arthritis
3. Enanthema
4. Cerebellar infarction

5. Superinfection bacterial
6. Dyslipidaemia
7. Gastric perforation
8. Cardiovascular insufficiency
9. Generalised bullous fixed drug eruption
10. Hyperlipidaemia
11. Gout
12. Cholestasis
13. Stevens-Johnson syndrome
14. Duodenal ulcer haemorrhage
15. Hepatitis acute
16. Dermatitis exfoliative generalised
17. Ischaemic stroke
18. Photosensitivity reaction
19. Haematoma muscle
20. Drug eruption
21. Drug reaction with eosinophilia and systemic symptoms
22. Pancreatitis acute
23. Renal impairment
24. Cholangitis
25. Blood creatine phosphokinase increased
26. Rhabdomyolysis
27. Agranulocytosis
28. Faecaloma
29. Eosinophil count increased

- 30. Drug-induced liver injury
- 31. Paralysis
- 32. Blood creatinine increased
- 33. Interstitial lung disease
- 34. Liver injury
- 35. Muscle atrophy
- 36. Musculoskeletal discomfort
- 37. Hyperkalaemia
- 38. Hepatic steatosis
- 39. Skin ulcer
- 40. Nephrolithiasis
- 41. Increased appetite
- 42. Acute kidney injury
- 43. Colitis
- 44. Hepatic function abnormal